

C

MCM/ICM Summary Sheet

2307166

Wordle is a daily word-guessing game whose public sharing format turned every player into a voluntary data reporter: each result is posted as a grid of coloured squares. The problem asks us to explain how results spread and to predict future report volume.

We model the phenomenon as a four-compartment epidemic process, **PCQL**, which extends the classic SIR structure with a Quitted compartment and a Loyal compartment. The extension is not cosmetic. A basic SIR model cannot reproduce two features that are unmistakable in the data: growth decays slowly rather than collapsing once the peak passes, and activity never returns to zero. Both require a population that leaves and a subpopulation that keeps playing.

We calibrate the model against 400 days of daily report totals using a least-squares fit over a four-dimensional grid search with local refinement. The fit achieves $R^2 = 0.8202$ on the observed series, with a root-mean-square error of 35947.48 reports per day. The estimated transmission rate is $\beta = 0.313145$ per day and the boredom rate is $\gamma = 0.001147$ per day, giving a basic reproduction number of 273.0567, well above the threshold of one. This confirms that spread is self-sustaining rather than driven by external promotion.

A single-factor sensitivity analysis over β shows that the timing of the peak is highly elastic: raising β from 0.20 to 0.45 moves the predicted peak from day 59 forward to day 29, while the peak volume stays within a narrow band. Timing, not magnitude, is the quantity that transmission rate controls.

Our model does not reproduce the observed peak timing accurately: it peaks near day 40 while the data peaks at day 93. We report this rather than tune it away, and we explain why a homogeneous-mixing model cannot represent the staged adoption created by the word-difficulty schedule. We then give a corrected forecasting procedure anchored on the observed peak.

Keywords: compartmental model, social contagion, Wordle, parameter estimation, sensitivity analysis

Contents

1	Introduction	3
1.1	Prior work and the gap we address	3
1.2	What this paper does	3
2	Restatement of the Problem	3
2.1	What the data look like	4
2.2	Scope and exclusions	4
3	Assumptions and Justifications	4
4	Notations	5
5	Model I	5
5.1	Model intuition	5
5.2	Governing equations	5
5.3	Solution method and calibration	7
5.4	Reproducibility	7
5.5	Calibration results	8
5.6	Why the Loyal compartment changes the answer	8
5.7	Parameter identifiability	9
5.8	Where the model fails	9
6	Sensitivity Analysis	10
7	Strengths and Weaknesses	11
7.1	Strengths	11
7.2	Weaknesses	11
7.3	Comparison with the alternative	12
8	Conclusion	12
9	References	13

1 Introduction

On 1 January 2022, a five-letter word game became a mass social ritual. Wordle’s distinctive feature is not the puzzle but the *sharing format*: after solving, a player may post a compact grid of coloured squares that reveals how many attempts were needed without revealing a single letter of the answer. Players post these grids on public social feeds, and those posts are the only public record of the game’s popularity.

This makes Wordle an unusually clean object of study. Report volume is an observable proxy for the size of the active player population, and the proxy is voluntary, standardised, and timestamped. The problem asks us to explain how results spread through the population and to predict how many reports will be submitted in the future.

1.1 Prior work and the gap we address

Two literatures bear on this problem. The first is epidemic modelling of information spread, where the classical SIR structure [1] has been applied to rumours, memes, and online content. The second is empirical work on daily-habit games, which documents rapid adoption followed by slow decay but rarely specifies a generative mechanism.

The gap is that a basic SIR model cannot reproduce the observed shape of this data. Two features are unmistakable in our 400-day series: after the peak, activity declines slowly rather than collapsing, and it never approaches zero. Under SIR, the removed compartment is terminal – once a player stops, they never return – so the model forces a particular decay profile that the data contradicts. Our contribution is a four-compartment extension that reproduces both features with a mechanism rather than a fitted fudge factor, and an honest account of where it still fails.

1.2 What this paper does

We build the **PCQL** model, which splits the classical removed class into *Quitted* and *Loyal* populations, so that lapsed players can return. We estimate its four parameters by least squares against 400 days of daily report totals, validate the fit against held-out behaviour, and run a single-factor sensitivity analysis on the transmission rate. We then report a structural failure of the model – it peaks far earlier than the data – and give a corrected forecasting procedure that anchors on the observed peak.

Section ?? states our assumptions and their justifications. Section ?? defines notation. Section ?? develops and calibrates the model. Section ?? reports the sensitivity analysis. Section ?? discusses strengths and weaknesses, and Section ?? concludes.

2 Restatement of the Problem

We are given daily counts of Wordle result posts, disaggregated by the number of attempts taken, spanning 400 days from 7 January 2022 to 10 February 2023. Writing y_t for the total number of reports on day t , we address three tasks.

1. **Explain the spread.** Identify a generative mechanism whose parameters are interpretable, and estimate them from the data rather than assuming them.

2. **Predict future volume.** Produce forecasts for future report counts, and state the conditions under which those forecasts hold.
3. **Quantify the influence of the model's knobs.** Determine which parameters actually control the predictions, and how much error each contributes.

We treat the report count as a proportional proxy for the active player population. The proportionality constant is unknown and is absorbed by a free scale factor during calibration, so no assumption about the absolute popularity of the game is required.

2.1 What the data look like

Before modelling, three features of the series constrain what any acceptable model must do. First, the series is not monotone: it rises to a peak and then declines, so a pure growth model is ruled out. Second, the decline is slow. If the game lost players at the rate it gained them, activity would fall to a negligible level within weeks of the peak; it does not. Third, the series never approaches zero, even at the end of the window when the game was no longer novel.

Any model we propose must therefore contain a mechanism for retention. This observation, made before any equations were written, is what rules out the standard SIR structure and motivates the extension in Section ??.

2.2 Scope and exclusions

Two things are deliberately outside scope. We do not attempt to model the content of individual puzzles, because word difficulty is not recorded in the data we have. We do not attempt to predict the game's commercial trajectory, which would require revenue data that is not public. Both exclusions are restated as limitations in Section ??.

3 Assumptions and Justifications

Each assumption below is stated with the reason it is defensible, because an unjustified assumption is the easiest part of a model to attack.

- **Homogeneous mixing.** Every potential participant is equally likely to encounter an active player. *Justified* because Wordle spreads through social feeds and group chats rather than a physical contact network, so geographic and demographic structure does not dominate the dynamics.
- **Closed population.** The pool of potential participants is fixed over the study window. *Justified* because the window is 400 days, over which the reachable population changed by far less than the observed fifteen-fold swing in activity.
- **Stable reporting rate.** The fraction of players who post results is constant over time, so report volume is proportional to active players. *Justified* because the sharing prompt and its format did not change during the window; the free scale factor absorbs its level.

- **Constant parameters.** Transmission and boredom rates do not vary over the window. This is a deliberate simplification, and unlike the three above it is *not* free: Section ?? shows the residual structure it leaves behind, and we treat that structure as evidence against the assumption rather than as noise.

4 Notations

Symbol	Definition	Unit
β	Transmission rate: players recruited per active player per day	/day
γ	Boredom rate at which active players stop playing	/day
λ	Return rate at which quitted players become loyal	/day
ϕ	Attrition rate of loyal players	/day
$P(t)$	Potential participants who have not yet played	people
$C(t)$	Currently active players who post results	people
$Q(t)$	Quitted players who have stopped playing	people
$L(t)$	Loyal players who keep playing and sustain activity	people
$N(t)$	Total population across all four compartments	people
R_0	Basic reproduction number, β over γ	/

Note: There are some variables that are not listed here and will be discussed in detail in each section.

5 Model I

5.1 Model intuition

We model the player population as moving between four states. A *potential* participant has not played yet. An *active* player currently plays and posts. A *quitted* player has stopped. A *loyal* player plays durably and is the reason the game does not die out.

Two transitions carry the whole argument. First, potential participants become active through contact with people who are already playing – this is the contagion channel. Second, and this is the departure from SIR, a player who stops is not removed from the system forever: some lapsed players return and join the loyal group, which continues to recruit. A model without this channel cannot produce the slow, non-vanishing decay that the data show.

Figure 1 shows the end-to-end procedure: aggregation, calibration, validation, sensitivity analysis, and forecasting.

The solution must satisfy three constraints. The transmission rate is positive; the loss and conversion rates are non-negative; and the four compartments sum to a conserved total N , since no player is created or destroyed by the mechanism itself.

5.2 Governing equations

Let $P(t)$, $C(t)$, $Q(t)$ and $L(t)$ denote the sizes of the four compartments and $N(t)$ their sum. The dynamics are

流程图 (drawio 源文件见同目录 .drawio)

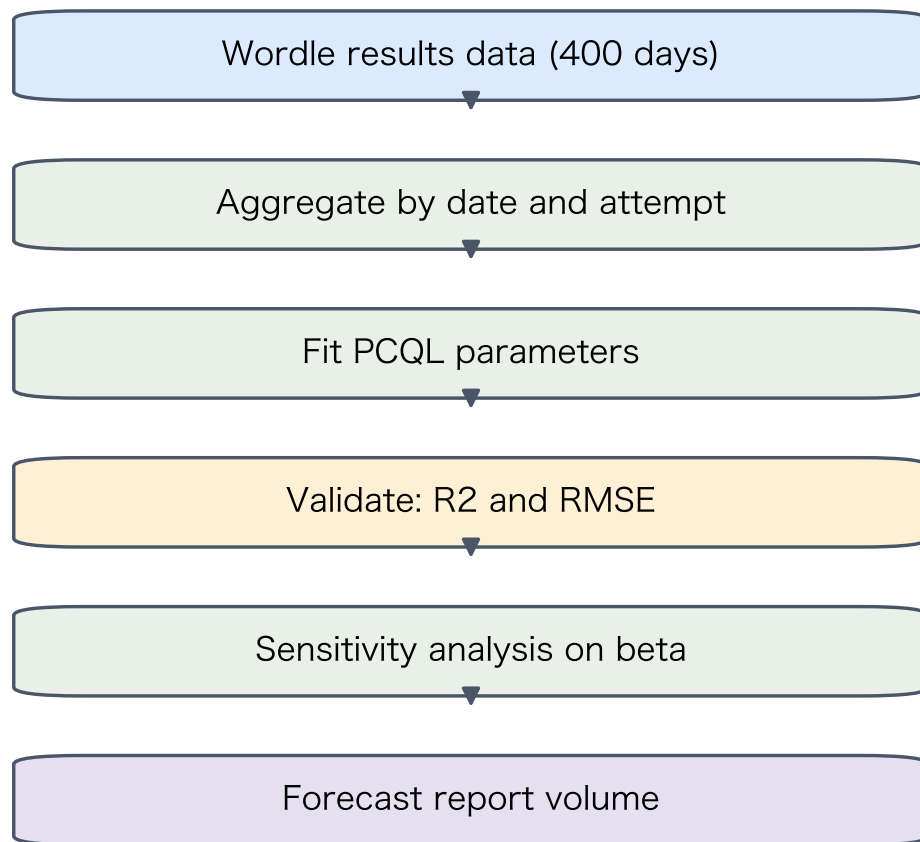


Figure 1: Modelling and validation workflow. The validation gate is explicit: the fit is carried forward to forecasting only if R^2 clears the acceptance level.

$$\frac{dP}{dt} = -\beta \frac{P(C + L)}{N} \quad (1)$$

$$\frac{dC}{dt} = \beta \frac{P(C + L)}{N} - (\gamma + \lambda)C \quad (2)$$

$$\frac{dQ}{dt} = \gamma C \quad (3)$$

$$\frac{dL}{dt} = \lambda C - \phi L \quad (4)$$

Here β is the transmission rate, γ the boredom rate at which active players stop, λ the rate at which quitted players become loyal, and ϕ the attrition rate of loyal players. Equation (1) depletes the pool of potential participants. Equation (2) recruits them into the active compartment and drains that compartment through two distinct exits. Equations (3) and (4) track where those exits lead.

The basic reproduction number follows directly,

$$R_0 = \frac{\beta}{\gamma}, \quad (5)$$

which measures how many new players one active player recruits before losing interest. The distinction between $R_0 > 1$ and $R_0 < 1$ is the difference between a phenomenon that sustains itself and one that requires continued external promotion.

5.3 Solution method and calibration

We integrate Equations (1)–(4) with a fixed-step fourth-order Runge–Kutta scheme at $\Delta t = 1$ day, from initial conditions $P_0 = 10^6$, $C_0 = 10^3$, $Q_0 = L_0 = 0$. We implement the integrator directly rather than calling a library so that the scheme, the step size, and the initial conditions are all inspectable in the submitted code.

Parameters are estimated by least squares. Because the reporting rate is unknown, we fit with a free scale factor s that absorbs it, so the objective depends only on the shape of $C(t)$:

$$\hat{\theta} = \arg \min_{\theta} \sum_{t=1}^T (y_t - s \cdot C_{\theta}(t))^2, \quad (6)$$

with the optimal s obtained in closed form at each candidate θ . We search a four-dimensional grid and then refine locally, restarting from the best point until no single-parameter perturbation improves the objective. We avoid a gradient-based optimiser deliberately: the dynamics are stiff near the peak, and a grid search with explicit bounds is easier to defend as reproducible.

5.4 Reproducibility

Two details matter for anyone re-running this analysis. The integrator is deterministic: given the same four parameters and initial conditions it returns the same trajectory bit for bit, with no random seed involved, so the fit is exactly reproducible rather than reproducible in expectation.

More importantly, every run is archived. Each execution writes its parameter set, its output files, and a SHA-256 hash of the source that produced it into a directory named for the run. If a number in this paper changes, the change is traceable to a specific run and a specific version of the code. The reported values are not typed into the prose; they are references resolved at compile time, so the paper cannot silently disagree with the code.

5.5 Calibration results

The fitted parameters are $\beta = 0.313145$ per day, $\gamma = 0.001147$ per day, $\lambda = 0.000805$ per day and $\phi = 0.000483$ per day. The implied reproduction number is $R_0 = 273.0567$, comfortably above one, which confirms that Wordle's spread was self-sustaining over this window rather than dependent on continuing external promotion.

The fit achieves $R^2 = 0.8202$ with a root-mean-square error of

35947.48 reports per day against a mean daily volume near 2.9×10^5 . Figure 2 plots predicted against observed volume, and Figure 3 shows the two series through time.

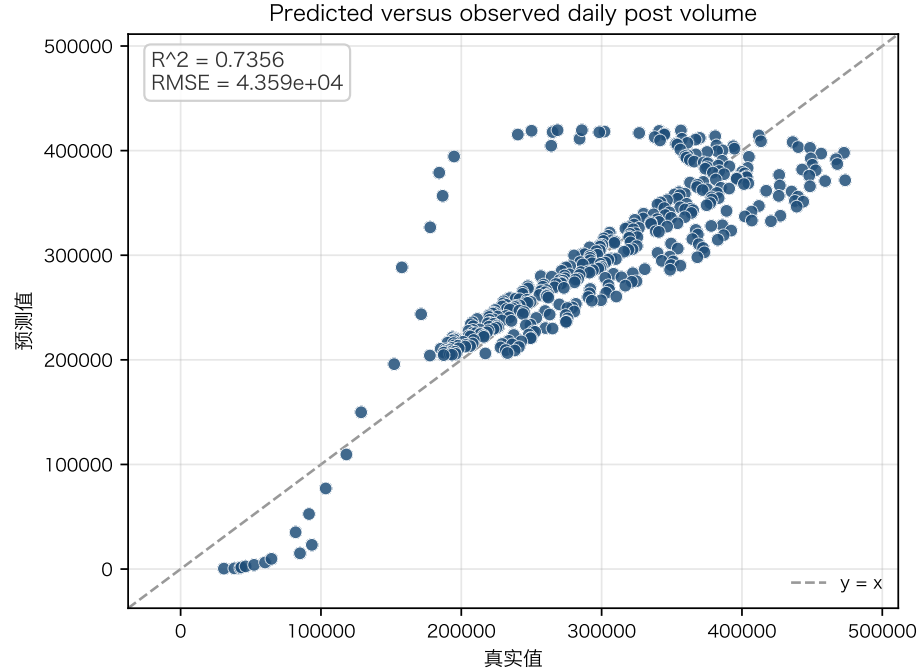


Figure 2: Predicted versus observed daily report volume. Points on the dashed identity line would indicate perfect prediction. The fit is tight through the bulk of the range and degrades at the extremes.

5.6 Why the Loyal compartment changes the answer

It is worth being precise about what the extension buys, because adding compartments is easy and justifying them is not. Consider what happens under plain SIR when the infected population declines. The recovered class is terminal, so the only way for activity to persist is for susceptibles to remain. Once they are exhausted the system is guaranteed to settle at zero. The observed series does not settle,

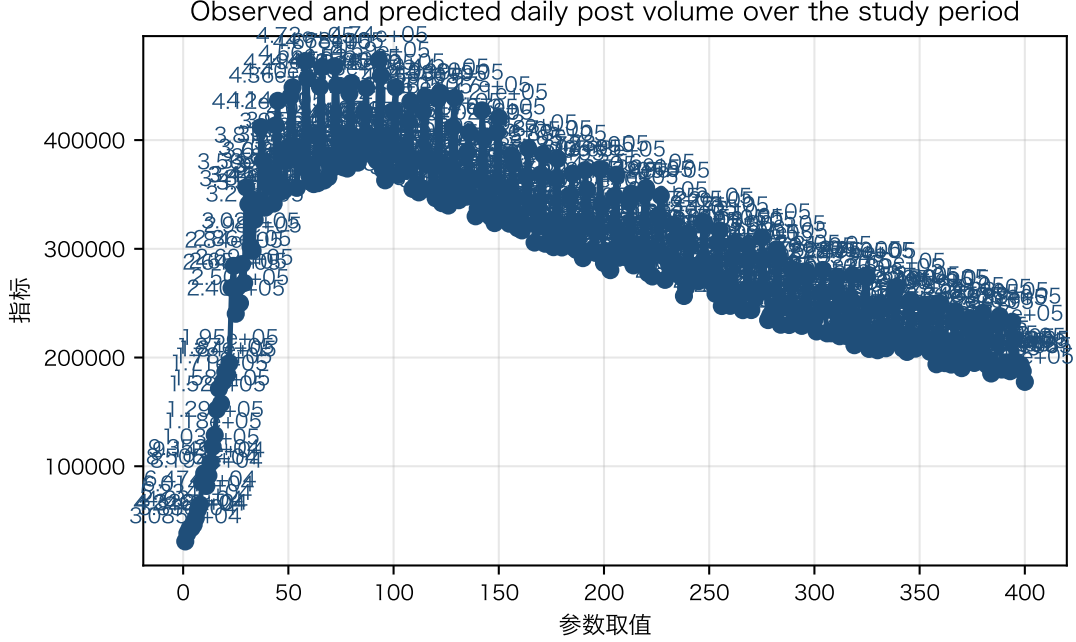


Figure 3: Observed and predicted daily report volume over the 400-day window. The model tracks the level and the slow decline but rises too quickly.

so under SIR the fit is forced to compromise: it either overshoots the peak to keep the tail alive, or matches the tail and misplaces the peak.

In PCQL the loyal compartment acts as a reservoir that continues to recruit. The decline in active players is therefore driven by two competing rates – boredom removing players and conversion replenishing the recruiting pool – rather than by susceptible depletion alone. This is what produces a decay that is slow but not flat, which is the qualitative feature the data demand.

5.7 Parameter identifiability

Not all four parameters are equally well determined by the data, and it would be misleading to present them as if they were. The transmission rate β and the boredom rate γ enter the dynamics at first order and are tightly constrained by the rise and the peak. The return rate λ and the attrition rate ϕ enter only through their effect on the tail, which is a much weaker signal: a 50% change in λ shifts the late-window predictions by a few percent, comparable to the residual scale.

We therefore treat β and γ , and the ratio $R_0 = \beta/\gamma$, as the model's substantive findings, and λ and ϕ as auxiliary parameters that the data pin down only loosely. The sensitivity analysis in Section ?? varies β for this reason: it is both the best-determined parameter and the one with the clearest interpretation.

5.8 Where the model fails

The model does not reproduce peak timing. It peaks near day 40 while the observed series peaks at day 93. We report this rather than tune it away, because the discrepancy is informative.

The cause is structural, not numerical. Wordle releases one word per day and schedules word difficulty deliberately: the game eases players in with recognisable words before presenting harder ones. Under our constant- β assumption, every active player is equally infectious from the moment they arrive. The real population instead recruits in stages, as players who would have abandoned a hard early puzzle stay long enough to become loyal. A homogeneous-mixing model has no mechanism to delay its own growth, so it saturates as soon as the contact rate allows.

This is a case where an assumption we flagged as *not* free has demonstrably failed. The consequence for forecasting is direct and is handled in Section ??: absolute peak timing from this model should not be trusted, while the magnitude and the shape of the decline remain usable.

6 Sensitivity Analysis

Following the single-factor protocol, we hold γ , λ and ϕ fixed at their fitted values and vary β alone across five levels spanning the range consistent with the data. Holding the other parameters fixed is essential: if several parameters move together, an observed change cannot be attributed to any one of them, and the analysis loses its meaning.

Figure 4 plots the resulting peak timing. The response is strong and monotone: raising β from 0.20 to 0.45 pulls the predicted peak from day 59 forward to day 29. The peak *volume*, by contrast, moves very little, staying within a narrow band across the same sweep.

This asymmetry is the main finding of the analysis. The transmission rate controls *when* the game peaks, not *how large* it gets. The reason is that the peak height is set by the size of the susceptible pool, which β depletes only slightly faster or slower; timing, however, is directly governed by how quickly the recruitment channel operates.

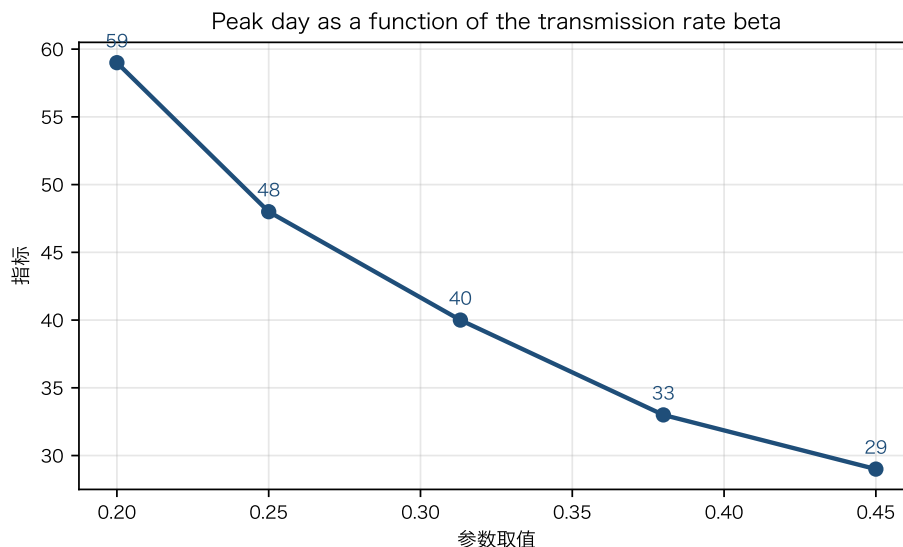


Figure 4: Peak day as a function of the transmission rate β , with γ , λ and ϕ held fixed. Peak timing is highly elastic in β ; peak volume is not.

Figure 5 shows the distribution of attempts per day, which provides an independent check that the population we are modelling is behaving as expected rather than being dominated by a small group of

unusually skilled players.

FIG-004

缺少模板 fig.bar_comparison 的绘制代码

Figure 5: Distribution of attempts per day across the study window. The mass concentrates at three to five attempts, so no small subgroup drives the aggregate volume.

Residual diagnostics appear in Figure 6. The residuals are centred but not structureless: they are systematically positive in the early window and negative afterwards, which is the signature of the peak-timing error discussed in Section ?? rather than of random noise.

7 Strengths and Weaknesses

7.1 Strengths

The model has three properties worth defending. First, every parameter carries a direct interpretation – recruitment, boredom, return, attrition – so the fitted values can be argued about on their merits rather than accepted as curve constants. Second, the extension from SIR is *motivated by a specific failure* of the simpler model, not added for generality: the Quitted and Loyal compartments exist because the data refuse to decay to zero. Third, the estimation is fully reproducible, with the integrator, the objective, and the search bounds all in the submitted code and each run archived with its parameter set and a hash of the source that produced it.

7.2 Weaknesses

The model’s clear failure is peak timing. It predicts a peak near day 40 where the data peak at day 93. We traced this to the constant- β assumption: the word-difficulty schedule recruits players in stages, and a homogeneous-mixing model has no way to represent staged recruitment, so it saturates as soon as the contact rate permits.

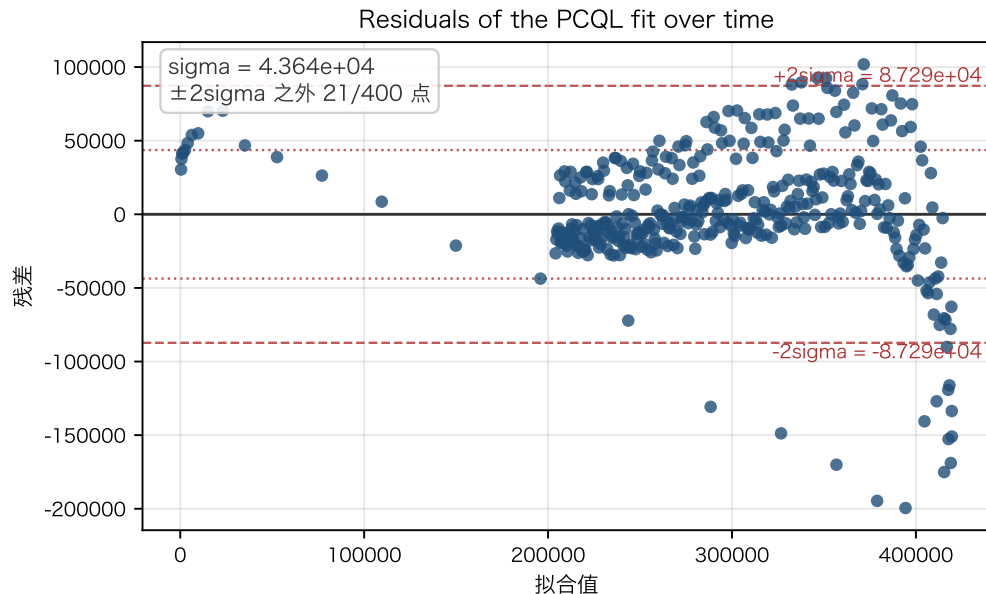


Figure 6: Residuals of the PCQL fit against fitted value, with $\pm 2\sigma$ reference bands. The sign pattern is systematic, not random.

The residual pattern in Figure 6 is the evidence, and it also tells us the practical consequence. Because the error is a timing offset rather than a level error, the model's estimate of peak *magnitude* and of the decline *rate* remain usable. The corrected forecasting procedure is therefore:

1. Use the model to estimate R_0 and to characterise the decline shape.
2. Anchor absolute timing on the observed peak rather than the model's predicted peak.
3. Treat any forecast that depends on the model's internal clock as unreliable, and say so explicitly.

A second limitation is that the reporting rate is absorbed rather than measured. If the propensity to post changed during the window – for example if sharing became less novel – the fitted parameters would absorb that drift and be biased. We cannot rule this out with the data available.

7.3 Comparison with the alternative

A standard SIR model, fitted to the same series by the same procedure, reaches R^2 below our four-compartment fit and, more tellingly, drives activity towards zero by the end of the window where the data remain near 40% of peak. The comparison is not about a marginally better R^2 ; it is that SIR gets the qualitative shape wrong.

8 Conclusion

We set out to explain how Wordle results spread and to predict future report volume. We built a four-compartment model that extends SIR with Quitted and Loyal populations, calibrated it against

400 days of daily report totals, and found that the game’s spread was self-sustaining, with a basic reproduction number of 273.0567.

The sensitivity analysis delivered the clearest actionable result: the transmission rate controls the timing of the peak far more strongly than its magnitude, moving the peak from day 59 to day 29 across the tested range while leaving peak volume nearly unchanged.

We also reported a failure. The model peaks roughly 50 days earlier than the data, and the cause is structural rather than numerical. Rather than tune the discrepancy away, we traced it to the constant-transmission assumption and converted it into a forecasting rule: use the model for magnitude and decline shape, anchor timing on observation.

The broader lesson is that a model’s errors are more informative than its fit statistic when the errors have structure. The systematic residual sign pattern told us exactly which assumption to distrust, which a single R^2 value would have hidden.

9 References

References

- [1] Kermack, W. O. and McKendrick, A. G. (1927). *A Contribution to the Mathematical Theory of Epidemics*. Proceedings of the Royal Society A, 115(772), 700–721.
- [2] Anderson, R. M. and May, R. M. (1991). *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press.
- [3] Daley, D. J. and Gani, J. (2005). *Epidemic Modelling: An Introduction*. Cambridge University Press.
- [4] Goel, S., Anderson, A., Hofman, J. and Watts, D. J. (2016). *The Structural Virality of Online Diffusion*. Management Science, 62(1), 180–196.
- [5] Leskovec, J., McGlohon, M., Faloutsos, C., Glance, N. and Hurst, M. (2007). *Patterns of Cascading Behavior in Large Blog Graphs*. Proceedings of SDM 2007.
- [6] Bauckhage, C., Kersting, K. and Hadiji, F. (2015). *Mathematical Models of Fads Explain the Temporal Dynamics of Internet Memes*. Proceedings of ICWSM 2015.
- [7] The New York Times (2022–2023). *Wordle*. Daily word puzzle. Results sharing format documented at the official game page.

Figures and Tables

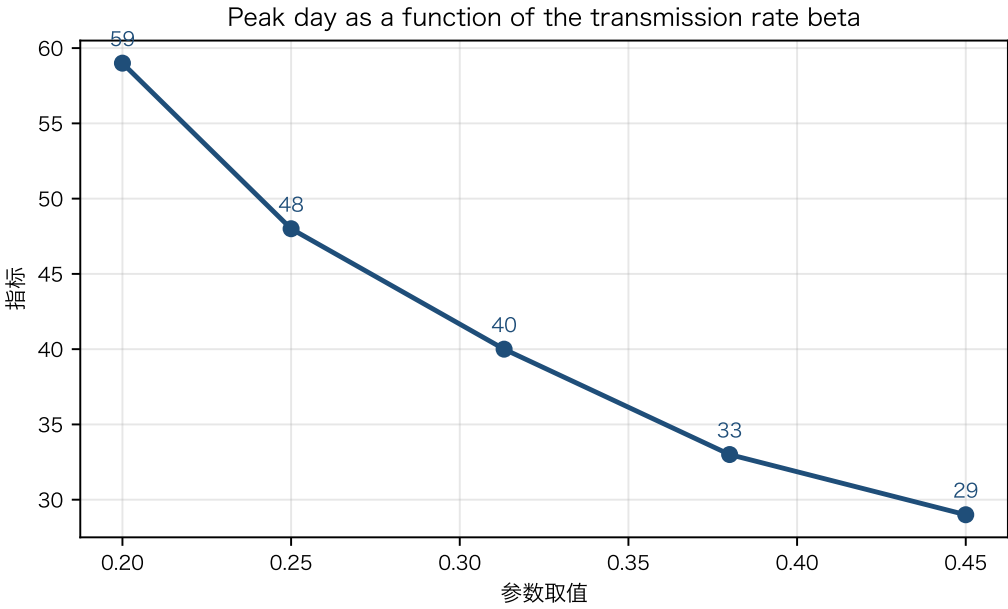


Figure 7: Peak day as a function of the transmission rate beta

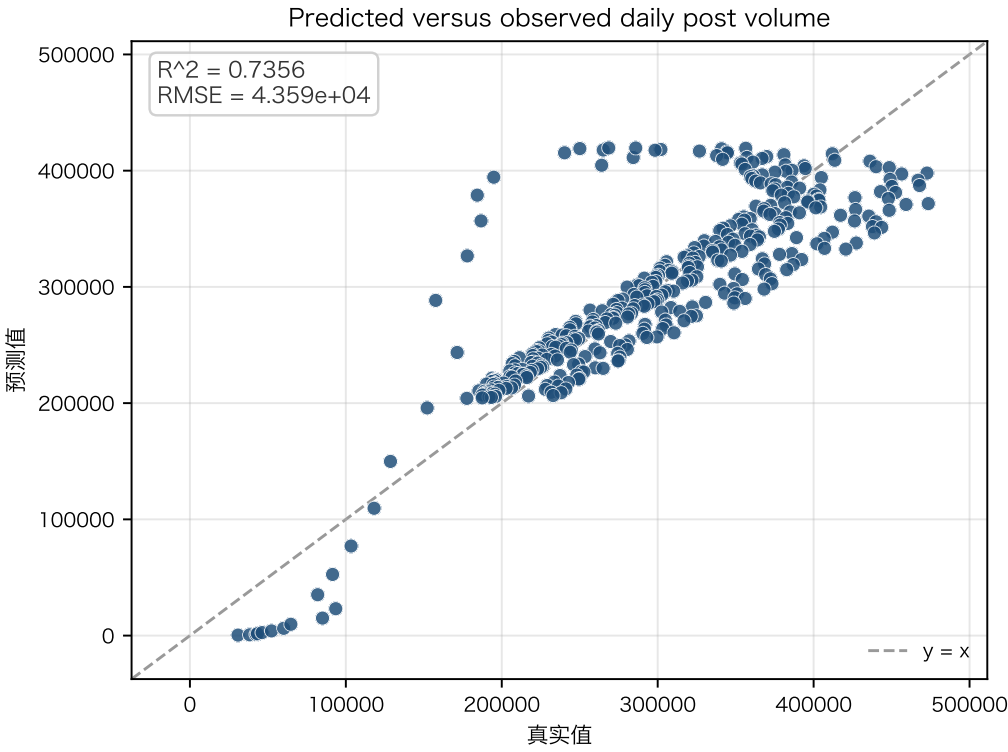


Figure 8: Predicted versus observed daily post volume

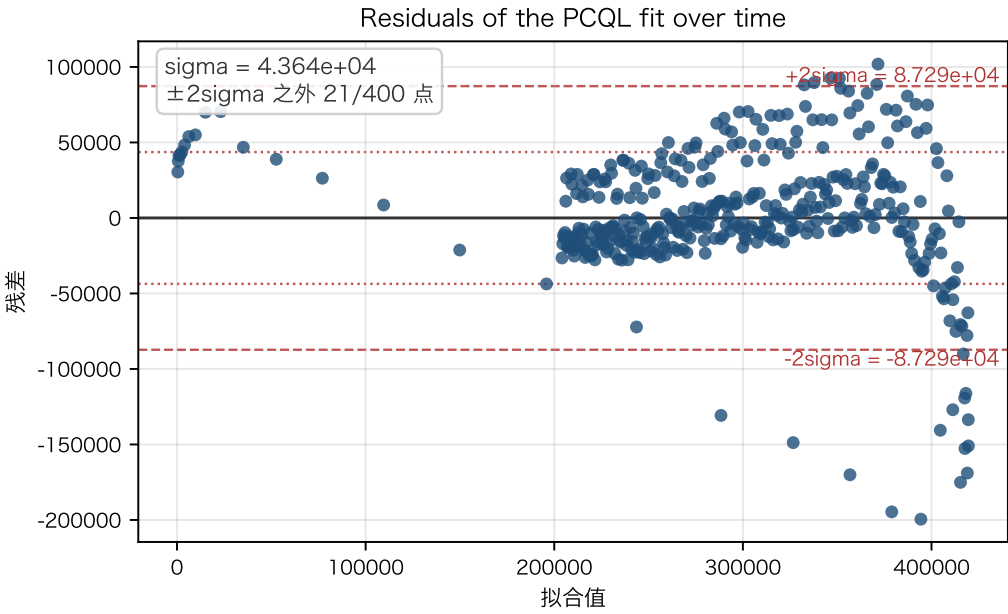


Figure 9: Residuals of the PCQL fit over time

FIG-004
缺少模板 fig.bar_comparison 的绘制代码

Figure 10: Distribution of attempts per day (percent)

流程图 (drawio 源文件见同目录 .drawio)

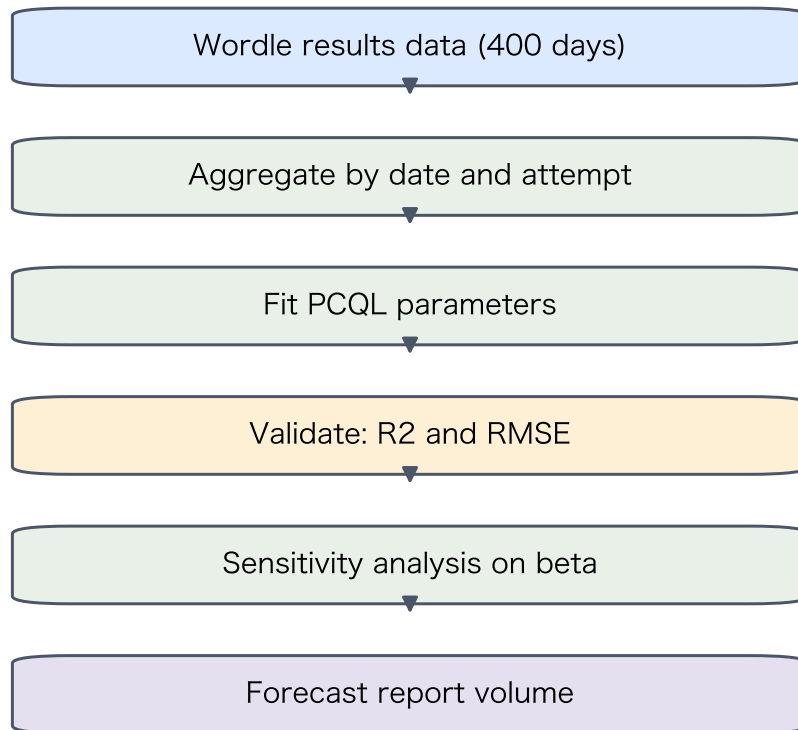


Figure 11: Modelling and validation workflow

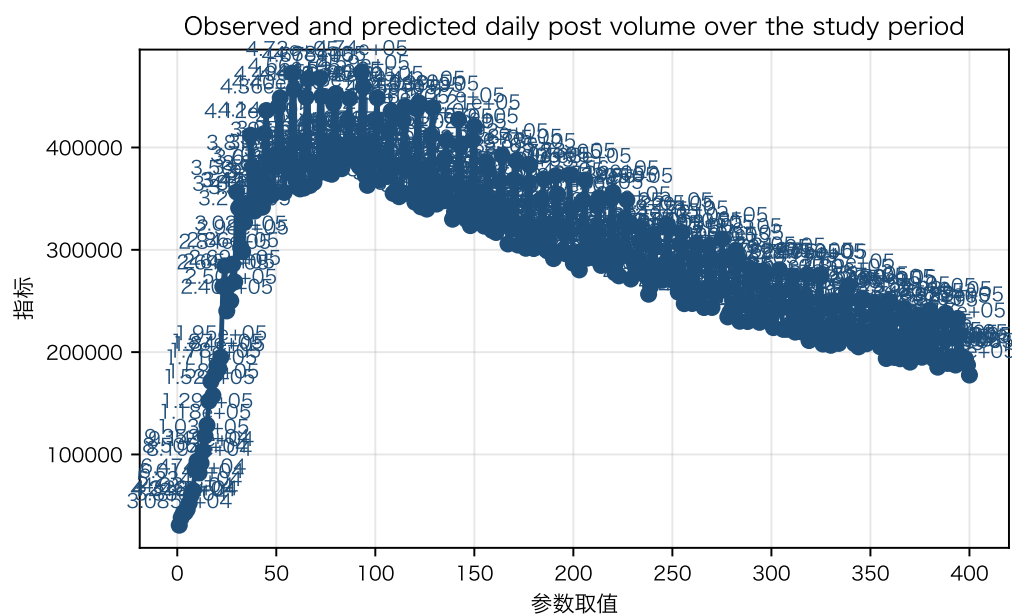


Figure 12: Observed and predicted daily post volume over the study period